Code No: 138GK **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July/August - 2022
MICROPROCESSORS AND MICROCONTROLLERS
(Common to CSE, IT)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Draw the internal architecture of 8086 microprocessor and explain.
b) Explain various addressing modes that the 8086 microprocessors can support. [7+8]

2.a) Write an 8086 Assembly language program to decide the parity of a given number.
b) Define memory segmentation and explain the concept of memory segmentation with diagram. [7+8]

3.a) Explain the memory organization of 8051 microcontroller.
b) Describe with examples various modes of the 8051 timers. [7+8]

4.a) Explain the different modes of serial communication in 8051.
b) Explain instruction set of 8051 microcontrollers. [7+8]

5.a) Write notes on RS-232 Serial data standard.
b) Explain interfacing of keyboard with 8051 microcontrollers using relevant diagram.[7+8]

6.a) Explain how to interface Analog -to- Digital Converter (ADC) to 8086 processor.
b) List out the steps involved in programming the 8051 to transfer data serially. [7+8]

7.a) Explain the generic structure of program status register as ARM core.
b) Explain the use of conditional branch instruction in Thumb Instruction Set. [7+8]

8.a) Explain the necessity of Architecture Revisions in ARM processor.
b) Explain the CORTEX processor with its architecture. [8+7]

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